Trolls, Haters, or Teammates: Classifying Sentiment in the Language of PvP Game Communications

UE24CS352A - Machine Learning Mini-Project | Vaishnavi (PES1UG24CS514) & Yuvraj Pateria (PES1UG24CS547)

1. Problem Statement

Real-time chat is essential in player-versus-player (PvP) games, but toxic messages drive players away, and standard sentiment tools fail on gamer slang: "gg" can be a friendly sign-off or a sarcastic taunt ("gg ez"). Following the Stanford CS229 report of the same name (Borst, Yang & Zhu, 2021), we classify each chat message into three classes: **Neutral / Positive**, **Negative attitude** (subtle hostility, mockery, blame) and **Derogatory** (explicit insults), using models fast enough for live moderation, and wrap the best one in an interactive demo.

2. Dataset

The report's authors hand-labelled 33k DOTA 2 and League of Legends chats; that data is not available to us, so we use the public **CONDA** dataset (Weld et al., ACL 2021) of human-annotated Dota 2 chat as the closest substitute. Its intent classes are mapped to the paper's scheme: explicit toxicity (E) → **Derogatory**, implicit toxicity (I) → **Negative attitude**, other chatter (O) → **Neutral / Positive**; ambiguous action requests (A) are dropped. This mapping is an approximation of the paper's labels. After cleaning and de-duplication within each split: train = 16,912 messages (Neutral/Positive 12,552, Negative 1,140, Derogatory 3,220); test = 5,949 (4,410 / 419 / 1,120). CONDA's official test split is unlabelled, so its validation file is our held-out test set.

3. Approach

Preprocessing: lower-casing, removing CONDA's [SEPA] markers, collapsing repeated characters, masking URLs. **Features:** (a) TF-IDF over word 1-2-grams plus character 2-5-grams (27,956 features), capturing slang phrases such as "gg ez" and typos; (b) the average of 50-d GloVe vectors per message. **Models:** K-Nearest Neighbours, Multinomial Naive Bayes, Logistic Regression, Linear SVM (probability-calibrated) and Random Forest on TF-IDF, and a feedforward neural network (MLP) on GloVe. Hyper-parameters were tuned on a stratified 15% validation split; final models were refit on all training data and scored once on the test set. The best model that outputs probabilities, by validation macro-F1, is deployed.

4. Implementation

Python / scikit-learn. `src/` contains the pipeline (data download, preprocessing, features, training, evaluation); `train_model.py` runs it end to end and saves `models/model.pkl` and `vectorizer.pkl`; `demo.py` is a Streamlit app showing the predicted class, class probabilities and the results table. An optional PyTorch Bi-LSTM script is included but was not run for this report.

5. Results

Model	Features	Accuracy	Macro-F1	Neg. attitude F1	ms / msg
K-NN	TF-IDF	0.856	0.695	0.51	29.2
Naive Bayes	TF-IDF	0.885	0.765	0.56	1.5
Logistic Regression	TF-IDF	0.897	0.799	0.59	1.4
Linear SVM	**TF-IDF**	**0.920**	**0.822**	**0.64**	**2.2**
Random Forest	TF-IDF	0.914	0.815	0.64	10.1
MLP + GloVe	GloVe-50	0.838	0.683	0.53	0.2

The deployed **Linear SVM** reaches **92.0% accuracy and 0.822 macro-F1** on unseen messages (~2.2 ms each). Random Forest is close but ~4x slower; the GloVe network (macro-F1 0.68) and K-NN trail. GloVe is trained on news text and covers only 87% of the chat tokens, so slang is invisible to it, supporting the paper's conclusion that frequency-based encodings suit gaming chat better than pre-trained embeddings. Per-class F1 for the deployed model: Neutral/Positive 0.95, Derogatory 0.87, Negative attitude 0.64.

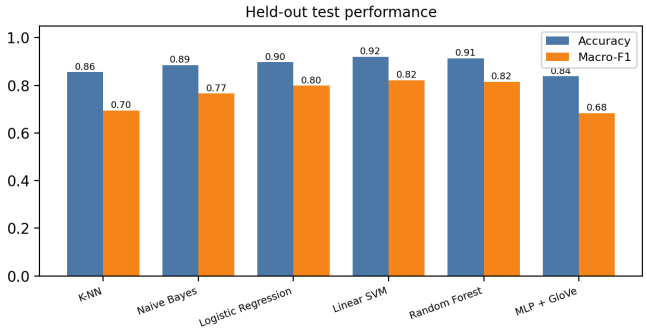


Fig. 1 - Accuracy and macro-F1 by model

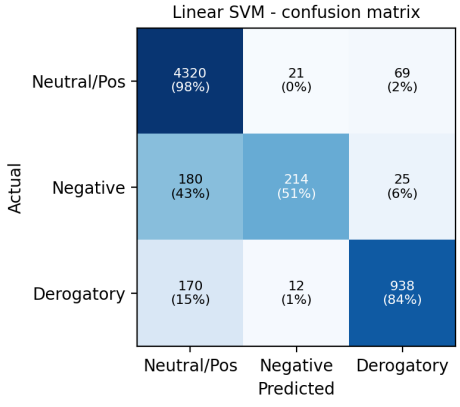


Fig. 2 - Linear SVM confusion matrix (counts, row %)

Error analysis. Negative attitude is the hard class: 43% of such messages are predicted as Neutral/Positive, because implicit hostility ("did you buy an account?") shares vocabulary with normal chatter and depends on context. Derogatory messages carry recognisable insults (the top learned features include ez, noob, trash, stupid, idiot). Some errors are annotation noise or sarcasm. **Leakage check:** 12.6% of test messages (mostly short ones like "gg") also occur in the training set; on the remaining novel messages the model scores 91.7% accuracy / 0.815 macro-F1, so the result is not inflated by duplicates.

6. Conclusions

Lightweight TF-IDF models, especially a calibrated linear SVM, classify PvP chat with 92% accuracy at millisecond latency, so they can screen every message in a live pipeline, and they outperform a GloVe-based neural network on slang-heavy text. Limitations: Dota 2 data only; labels are an approximation of the paper's; context and sarcasm are not modelled; Negative attitude is under-detected. **Future work:** use preceding messages as context, fine-tune a transformer, and gather more labelled negative-attitude examples.